The impulsive close–encounter cutoff is exactly $b_{90} = G(m_i + m_*)/V^2$: a self–contained proof

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Abstract

We prove that, for a point–mass perturber, the ultraviolet (close–encounter) divergence of the impulsive kick covariance of a binary is regularized by the 90°–deflection radius $b_{90,i} = G(m_i + m_*)/V^2$ *with no free constant*, and that the resulting Coulomb logarithm and the accompanying $\mathcal{O}(1)$ constants are the exact leading–order result in the impulsive limit $\eta = b_{90}/r \to 0$. The proof combines the exact two–body (Rutherford) kick, a clean single–body self–energy identity, an exact variation–of–parameters (osculating–elements) treatment of the close encounter perturbed by the companion, and a matched decomposition of the impact–parameter plane whose intermediate scale cancels identically. Every step is reduced either to an explicit computation or to elementary quantitative estimates (Gronwall bounds on the incoming leg, an angular–momentum bound for close approaches, measure estimates for sequential encounters); the only inputs kept outside the argument are the frozen–binary approximation itself and Newtonian gravity. Two physical points deserve emphasis. First, the recoil of the struck body during the encounter is essential: pinning the binary components in place would change the cutoff to Gm_i/V^2; it is the recoil that produces the reduced–mass scale $G(m_i + m_*)/V^2$. Second, the companion deflects the perturber by an $\mathcal{O}(1)$ amount *per trajectory* even in the limit; this survives the impact–parameter integration only as an $\mathcal{O}(\eta)$ correction (numerically consistent with a gravitational–focusing effect), because to leading order it is an area–preserving relabeling of the impact–parameter plane. The proven error bound is a positive power of η; the numerics show the true rate is $\mathcal{O}(\eta)$ with no constant offset, directly excluding any rescaling $b_{90,i} \to \alpha\, b_{90,i}$.

Contents

This note accompanies the paper "Evolution of Binaries Under Stochastic Perturbations", where the result proven here is used; notation follows that paper. The numerical checks quoted below are the scripts `exact_deflection_check.py`*,* `three_body_check.py` *and* `deflection_map_check.py`*, released alongside this note in the same directory (*`regularization-exactness/`*) of the paper's code repository; the companion script* `cutoff_check.py` *additionally documents the Fourier-space origin of the two per-body cutoffs and the scheme dependence of alternative regulators.*

1 Setup, conventions, and the precise claim

1.1 The encounter: the frozen–binary model, stated carefully

A binary has components of masses m_1, m_2 (total $m = m_1 + m_2$) initially at rest at positions $\vec{s}_1, \vec{s}_2$ with separation $\vec{r} = \vec{s}_2 - \vec{s}_1$, $|\vec{r}| = r$. A perturber of mass m_* arrives from infinity with velocity $\vec{V}$ ($|\vec{V}| = V$) and asymptotic impact parameter $\vec{b}$ lying in the plane Π_V through the centre of mass perpendicular to $\hat{V} = \vec{V}/V$.

The dynamics is Newtonian gravity among the three bodies *with the internal binary force switched off*: the perturber is accelerated by both components, each component is accelerated by the perturber, and the components do not act on each other,

$$\ddot{\vec{x}}_* = -\sum_{i=1,2} \frac{Gm_i(\vec{x}_* - \vec{x}_i)}{|\vec{x}_* - \vec{x}_i|^3}, \qquad \ddot{\vec{x}}_i = \frac{Gm_*(\vec{x}_* - \vec{x}_i)}{|\vec{x}_* - \vec{x}_i|^3} \quad (i = 1, 2). \tag{1}$$

This is the impulse approximation: the internal force is what makes the binary orbit on the slow time scale, and freezing it isolates the effect of one fast passage; its leading correction (the binary moving during the passage) is the standard adiabatic correction, of relative order $Gm/(rV^2) = \mathcal{O}(\eta)$ [1], the same order as the corrections we track, and is outside the scope of this note (assumption (A1) below). The net kicks are $\Delta\vec{v}_i \equiv \vec{v}_i(+\infty) - \vec{v}_i(-\infty) = \vec{v}_i(+\infty)$, and the change of the binary's internal (relative) velocity is $\Delta\vec{v} = \Delta\vec{v}_1 - \Delta\vec{v}_2$ (this is minus the change of $\mathrm{d}(\vec{s}_2 - \vec{s}_1)/\mathrm{d}t$; the sign is irrelevant for the second moments studied here).

Remark 1 (Recoil is essential: pinned bodies would give a different cutoff)**.** *One might be tempted to freeze the binary by* pinning *the components, computing the perturber's orbit in the fixed two–centre potential and reading off the would–be impulse* $\Delta\vec{v}_i = Gm_* \int (\vec{x}_* - \vec{s}_i)|\vec{x}_* - \vec{s}_i|^{-3}\mathrm{d}t$*. That model gives a* different *answer. For a single pinned body the perturber's orbit is a hyperbola with gravitational parameter* Gm_i *(the source does not participate in the reduced–mass dynamics), so*

$\tan(\Theta/2) = \tilde{b}_i/\beta$ *with* $\tilde{b}_i = Gm_i/V^2$, *and by momentum bookkeeping* $|\Delta\vec{v}_i|^2 = 4G^2m_*^2/[V^2(\beta^2+\tilde{b}_i^2)]$*: the close–encounter cutoff would be* Gm_i/V^2, *not* $G(m_i+m_*)/V^2$, *an error of* $\ln[(m_i+m_*)/m_i]$ *in the Coulomb–log constant (vanishing only for* $m_* \ll m_i$*). In the correct model* (1) *the struck body recoils during the passage — by a displacement* $\mathcal{O}(b_{90})$, *the same order as the cutoff itself — and the relative* m_**–*m_i *motion is the reduced–mass Kepler problem with parameter* $G(m_i+m_*)$, *which produces the literal* $b_{90,i}$ *of the theorem. Both statements are confirmed by direct integration in* `deflection_map_check.py` *(§7.2).*

We use throughout the 90°–deflection radii and the small parameter

$$b_{90,i} \equiv \frac{G(m_i+m_*)}{V^2}, \qquad b_{90} \equiv \frac{G(m+m_*)}{V^2}, \qquad \boxed{\eta \equiv \frac{b_{90}}{r} = \frac{G(m+m_*)}{rV^2}}. \tag{2}$$

The *impulsive limit* is $\eta \to 0$, realized e.g. by $V \to \infty$ at fixed binary. All of $b_{90,i}, b_{90}$ differ only by bounded mass factors; η is their common dimensionless smallness. Throughout, $C, c, \ldots$ denote positive constants depending only on the mass ratios (never on η), whose value may change between displays.

1.2 The kick covariance

The local second–moment (diffusion) tensor of a bath of perturbers, number density n and isotropic velocity distribution g, is

$$\mathcal{Q}^{\alpha\beta}(r) = n \int \mathrm{d}^3V\, V\, g(\vec{V}) \int_{\Pi_V} \mathrm{d}^2b\, \Delta v^\alpha\, \Delta v^\beta = \mathrm{diag}(Q_r, Q_t, Q_t), \tag{3}$$

the diagonal form following from isotropy of g ($\hat{r}$ being the only special direction). For a *point* perturber the inner d^2b integral is logarithmically ultraviolet divergent in the straight–line approximation: as $\vec{b}$ brings the perturber arbitrarily close to a body, the straight–line kick $\propto 1/b$ blows up. The physical content of this note is the precise value of the scale at which the exact dynamics (1) cuts off this divergence.

1.3 The closed form to be proved

Fix $\vec{V}$ and do the d^2b integral first. Writing $\vec{y} \equiv (V/2Gm_*)\,\Delta\vec{v}$ and resolving the in–plane second moment along ($\perp$) and across ($\|$) the projected separation $\vec{r_\perp} = \vec{r} - (\vec{r}\cdot\hat{V})\hat{V}$ ($r_\perp = r\sin\theta$, $\cos\theta = \hat{V}\cdot\hat{r}$), the companion paper's result is the in–plane tensor

$$\boxed{\mathcal{Y}_\perp + \mathcal{Y}_\| = 2\pi \ln\frac{r_\perp^2}{b_{90,1}b_{90,2}}, \qquad \mathcal{Y}_\perp - \mathcal{Y}_\| = 2\pi} \tag{4}$$

(equivalently $\mathcal{Y}_\perp = \pi[\,\bar{L}+1\,]$, $\mathcal{Y}_\| = \pi[\,\bar{L}-1\,]$ with $\bar{L} = \ln[r_\perp^2/(b_{90,1}b_{90,2})]$), which after the elementary θ– and speed–averages (§6) gives

$$Q_r = \tfrac{8\pi}{3}C_Q\big(\mathcal{L} - \tfrac{2}{3}\big), \quad Q_t = \tfrac{8\pi}{3}C_Q\big(\mathcal{L} - \tfrac{8}{3}\big), \quad C_Q = G^2m_*^2n\langle V^{-1}\rangle, \tag{5}$$

with $\mathcal{L} = \ln\big[16\sigma^4r^2/(G^2(m_1+m_*)(m_2+m_*))\big] - 2\gamma_\mathrm{E}$ for a Maxwellian of dispersion σ. The boxed scales in (4) — the *literal* $b_{90,i}$ with unit coefficient inside the log, and the cutoff–free anisotropy constant 2π — are what fix the $\mathcal{O}(1)$ constants $-\frac{2}{3}, -\frac{8}{3}$ and the argument of $\mathcal{L}$. They are the object of the theorem.

Theorem 1 (Exactness of the impulsive regularization)**.** *Let $\mathcal{Y}_\perp^{\rm exact}, \mathcal{Y}_\parallel^{\rm exact}$ be computed from the exact frozen–binary three–body dynamics (1) for a point perturber. Fix any $\alpha \in (0, \frac{1}{2})$. Then, at fixed V, uniformly over orientations with $\sin\theta \geq \eta^\alpha$,*

$$\mathcal{Y}_\perp^{\rm exact} + \mathcal{Y}_\parallel^{\rm exact} = 2\pi \ln \frac{r_\perp^2}{b_{90,1} b_{90,2}} + o(1), \qquad \mathcal{Y}_\perp^{\rm exact} - \mathcal{Y}_\parallel^{\rm exact} = 2\pi + o(1), \tag{6}$$

where $o(1) \to 0$ as $\eta \to 0$ (indeed $\mathcal{O}(\eta^c)$ for an explicit $c > 0$; see §6.1). The near–aligned cone $\sin\theta < \eta^\alpha$ contributes $\mathcal{O}\big(\eta^{2\alpha} \ln^3(1/\eta)\big)$ to the θ–averages. Consequently, for a Maxwellian bath the constants are exact in the joint impulsive limit $Gm/(r\sigma^2) \to 0$:

$$\tfrac{3}{8\pi}\, Q_r^{\rm exact}/C_Q - \mathcal{L} \;\longrightarrow\; -\tfrac{2}{3}, \qquad \tfrac{3}{8\pi}\, Q_t^{\rm exact}/C_Q - \mathcal{L} \;\longrightarrow\; -\tfrac{8}{3}. \tag{7}$$

In particular no $\mathcal{O}(1)$ multiplicative ambiguity in $b_{90,i}$, and no additive constant in the log, survives: the regularization scale is exactly $b_{90,i} = G(m_i + m_*)/V^2$.

The restriction $\sin\theta \geq \eta^\alpha$ in the pointwise displays is necessary: for $\sin\theta \lesssim \sqrt{\eta}$ the two close–encounter regions overlap and (6) cannot hold uniformly. The averaged statement (7) is unconditional. The error term is numerically $\mathcal{O}(\eta)$ (§7); the weaker rate we prove is all that the constants require.

1.4 Assumptions and scope

We are explicit about the logical status of each ingredient.

(A1) **Frozen binary.** The model is (1): the internal binary force is switched off during the encounter, but the bodies are otherwise fully dynamical (they recoil). This is essential — see Remark 1. The leading correction to (A1) (the binary's internal motion during the passage) is the adiabatic correction, of relative order $\mathcal{O}(\eta)$, independent of the regularization question and equally vanishing in the limit.

(A2) **Newtonian point perturber.** No relativity; $\tilde{\rho}_*(k) = 1$. The whole question is meaningful only here, since any extended profile is self–regularizing.

(A3) **Rigour.** Lemmas 1–3 (single body), 4–6 (decomposition and measure estimates), 7–8 (near region), 9 (far region) and 10 (speed average) are proved. The proofs of Lemmas 7–8 reduce the dynamics to (i) an exact variation–of–parameters identity for the osculating elements of the m_*–m_1 relative orbit, and (ii) quantitative regularity of the *incoming leg* of the Kepler flow, stated as Lemma 11 in Appendix A with an explicit–formula proof strategy; the constants there are computable but not optimized. No scaling claim is used in place of a needed bound.

2 The exact single–body encounter

Everything close–range reduces to one exactly solvable problem: a perturber and *one* recoiling body. We record its kick, its self–energy, and the symmetry that lets the longitudinal recoil be carried by the trace.

Lemma 1 (Exact two–body kick)**.** *In an isolated Newtonian encounter of the perturber m_* with a single free body m_i, relative speed V at infinity and impact parameter β, the velocity change of body i has magnitude*

$$|\Delta\vec{v}_i|^2 = \frac{4G^2 m_*^2}{V^2} \frac{1}{\beta^2 + b_{90,i}^2}, \qquad b_{90,i} = \frac{G(m_i + m_*)}{V^2}, \tag{8}$$

exactly, for all $\beta \ge 0$. Resolving relative to the incoming direction $\hat V$,

$$|\Delta\vec v_i|_\perp^2 = \frac{4G^2m_*^2}{V^2}\frac{\beta^2}{(\beta^2+b_{90,i}^2)^2}\quad(\perp \hat V),\qquad |\Delta\vec v_i|_\parallel^2 = \frac{4G^2m_*^2}{V^2}\frac{b_{90,i}^2}{(\beta^2+b_{90,i}^2)^2}\quad(\parallel \hat V). \tag{9}$$

Proof. The relative orbit is a Kepler hyperbola with gravitational parameter $\mu_i = G(m_i + m_*)$; the relative velocity rotates by the scattering angle Θ with $\tan(\Theta/2) = b_{90,i}/\beta$ (standard, e.g. Rutherford). Since $|\vec V|$ is conserved, $|\Delta\vec V_{\rm rel}|^2 = 2V^2(1-\cos\Theta) = 4V^2\sin^2(\Theta/2)$. Using $\sin^2(\Theta/2) = \tan^2(\Theta/2)/[1+\tan^2(\Theta/2)] = b_{90,i}^2/(\beta^2+b_{90,i}^2)$ gives $|\Delta\vec V_{\rm rel}|^2 = 4V^2b_{90,i}^2/(\beta^2+b_{90,i}^2)$. The subject's change is $\Delta\vec v_i = \frac{m_*}{m_i+m_*}\Delta\vec V_{\rm rel}$ (centre–of–mass split), and $\frac{m_*}{m_i+m_*}Vb_{90,i} = Gm_*/V$, whence (8). For the resolution, the relative–velocity rotation by Θ contributes a transverse part $\propto \sin\Theta$ and a longitudinal part $\propto (1-\cos\Theta)$; $\sin^2\Theta = 4\beta^2b_{90,i}^2/(\beta^2+b_{90,i}^2)^2$ and $(1-\cos\Theta)^2 = 4b_{90,i}^4/(\beta^2+b_{90,i}^2)^2$ give (9), whose sum is (8). Note the reduced–mass parameter $\mu_i = G(m_i+m_*)$ enters only because the body recoils; cf. Remark 1. □

Remark 2. *The transverse part of* (9) *is identical to the straight–line impulse from a Plummer sphere of core $\beta_c = b_{90,i}$; the longitudinal part is the forward focusing that the straight–line picture omits. It is precisely this longitudinal piece that distinguishes the genuine deflection from a soft core, and it is what makes the next lemma land on the bare $b_{90,i}$ rather than $b_{90,i}\sqrt{e}$.*

Lemma 2 (Single–body self–energy)**.** *Let $b > 0$ be a core scale. For any cutoff $\Lambda \gg b$,*

$$\int_{\beta<\Lambda}\frac{\mathrm d^2\beta}{\beta^2+b^2} = \pi\ln\frac{\Lambda^2+b^2}{b^2} = 2\pi\ln\frac{\Lambda}{b} + \mathcal O\big(b^2/\Lambda^2\big)\,. \tag{10}$$

This equals the sharp–cutoff value $\int_b^\Lambda 2\pi\beta\,\mathrm d\beta/\beta^2 = 2\pi\ln(\Lambda/b)$ with the same *additive constant (namely none). Applied with $b = b_{90,i}$, and equivalently: the gravitational momentum–transfer cross section is $\sigma_{\rm t} = \int(1-\cos\Theta)\,\mathrm d\sigma = 4\pi b_{90,i}^2\ln(\Lambda/b_{90,i})$.*

Proof. Substitute $u = \beta^2$: $\int_0^{\Lambda^2}\pi\,\mathrm du/(u+b^2) = \pi\ln[(\Lambda^2+b^2)/b^2]$. Expanding for $\Lambda \gg b$ gives the stated form. The sharp–cut integral is elementary; both produce $2\pi\ln(\Lambda/b)$ with zero constant. The cross–section identity follows from $\mathrm d\sigma = 2\pi\beta\,\mathrm d\beta$ and (8) (or directly from the Rutherford $\mathrm d\sigma/\mathrm d\Omega$). □

The content of Lemma 2 is that the *exact* deflection regularizes the self–energy at the bare $b_{90,i}$. Had one used only the transverse part of (9) (the "soft core") one would get $\int\beta^2(\beta^2+b^2)^{-2}\mathrm d^2\beta = 2\pi\ln(\Lambda/b) - \pi = 2\pi\ln(\Lambda/(b\sqrt{e}))$; the longitudinal part supplies exactly the compensating $+\pi$ ($\int b^2(\beta^2+b^2)^{-2}\mathrm d^2\beta = \pi$), restoring the bare b. This is the entire "$\sqrt{e}$" scheme question, settled by including the genuine recoil.

Lemma 3 (Isotropy of the close contribution)**.** *Fix the close–encounter region $\beta < \rho$ (for any ρ) around a* single *body. At fixed $\vec V$ the single–body contribution $q^{\alpha\beta}(\hat V) \equiv \int_{\beta<\rho}\Delta v_i^\alpha\Delta v_i^\beta\,\mathrm d^2\beta$ is axially symmetric about $\hat V$,*

$$q^{\alpha\beta}(\hat V) = A\,\hat V^\alpha\hat V^\beta + B\,(\delta^{\alpha\beta} - \hat V^\alpha\hat V^\beta), \tag{11}$$

and therefore its average over isotropic $\hat V$ is isotropic, $\langle q^{\alpha\beta}\rangle_{\hat V} = \frac13\,\mathrm{tr}(q)\,\delta^{\alpha\beta}$. Consequently the close contribution to $\mathcal Q^{\alpha\beta}$ carries no anisotropy*: it splits its trace (which, by Lemma 1, already contains the longitudinal recoil) equally between Q_r and Q_t.*

Proof. For a single body the only distinguished direction is $\hat{V}$; integrating $\Delta\vec{v}_i$ (which lies in the plane of $\hat{V}$ and $\hat{\beta}$) over the azimuth of $\vec{\beta}$ about $\hat{V}$ yields a tensor diagonal in any frame adapted to $\hat{V}$, i.e. the displayed axially symmetric form. Averaging with $\langle \hat{V}^\alpha \hat{V}^\beta \rangle = \frac{1}{3}\delta^{\alpha\beta}$ gives $\frac{1}{3}(A+2B)\delta^{\alpha\beta} = \frac{1}{3}\,\mathrm{tr}(q)\delta^{\alpha\beta}$. □

Lemma 3 is what licenses the in–plane bookkeeping of (4): the trace (where the cutoff lives) may be computed including the longitudinal recoil, while the anisotropy comes *entirely* from the two–body interference (§5). Equivalently: at fixed $\hat{V}$, the difference between the exact single–body tensor and the paper's transverse sharp–b_{90} tensor is the traceless quadrupole $\pi\ \mathrm{diag}(-\frac{1}{2},-\frac{1}{2},+1) = \frac{3\pi}{2}(\hat{V}\hat{V} - \frac{1}{3}\mathbb{1})$, whose isotropic average vanishes and whose Q_r projection integrates to $\int_0^\pi \sin\theta[\pi\cos^2\theta - \frac{\pi}{2}\sin^2\theta]\mathrm{d}\theta = 0$. In the exact three–body dynamics the near tensor is axisymmetric only up to the map distortions bounded in §4; those errors are booked in the tally of §6.1.

3 Decomposition of the binary integral

Fix $\vec{V}$. Write the relative kick as $\Delta\vec{v} = \Delta\vec{v}_1 - \Delta\vec{v}_2$, so that the in–plane second moment involves

$$\int_{\Pi_V} \mathrm{d}^2 b\, |\vec{y}|^2 = \underbrace{\int |\vec{y}_1|^2}_{\text{self 1}} + \underbrace{\int |\vec{y}_2|^2}_{\text{self 2}} - \underbrace{2\int \vec{y}_1\cdot\vec{y}_2}_{\text{cross}}, \qquad \vec{y}_i \equiv \frac{V}{2Gm_*}\Delta\vec{v}_i. \tag{12}$$

Each self term carries one logarithm, divergent (in the straight–line approximation) both at small b (close encounter) and large b; the large–b divergences of the two self terms cancel against the cross term (the dipole cancellation of §5); the anisotropy $\mathcal{Y}_\perp - \mathcal{Y}_\parallel$ resides in the interference only (Lemma 3).

To separate the close–encounter scale (b_{90}) from the binary scale (r) cleanly, fix an intermediate radius

$$\rho \equiv \sqrt{b_{90}\, r}, \qquad b_{90} \ll \rho \ll r \quad (\text{indeed } \rho/r = b_{90}/\rho = \sqrt{\eta}), \tag{13}$$

and partition Π_V into the two *near* disks $\{\beta_i < \rho\}$ about the projected body positions ($\beta_i \equiv |\vec{b} - \vec{s}_i^{\,\mathrm{proj}}|$) and the *far* remainder $\{\beta_1 \geq \rho$ and $\beta_2 \geq \rho\}$. Orientations are split into the *generic* class $\sin\theta \geq \eta^\alpha$ (where $r_\perp \geq r\eta^\alpha \gg 2\rho$, so the near disks are far apart) and the *near–aligned cone* $\sin\theta < \eta^\alpha$, handled wholesale by Lemma 6. We fix $\alpha \in (0, \frac{1}{2})$ once and for all.

Two facts about the near disks, both needed later, are worth isolating. The first is an exact cancellation that the naive bound misses; the second is the measure estimate that replaces a pointwise bound which is *false* for the exact dynamics.

Lemma 4 (Harmonicity cancellation on the near disk)**.** *Let $\vec{u}_a(\vec{b}) = (\vec{b} - \vec{s}_a^{\,\mathrm{proj}})/\beta_a^2$ be the straight–line single–body kicks. Then, exactly,*

$$\int_{\beta_1<\rho} \vec{u}_1\cdot\vec{u}_2\,\mathrm{d}^2 b = 0 \qquad (\textit{and symmetrically on the disk around body 2}), \tag{14}$$

for any $\rho < r_\perp$. Moreover $\int_{\beta_1<\rho} |\vec{u}_2|^2\,\mathrm{d}^2 b \leq \pi\rho^2/(r_\perp - \rho)^2$.

Proof. $\vec{u}_a = \nabla \ln\beta_a$. On the disk $D = \{\beta_1 < \rho\}$ the function $\ln\beta_2$ is harmonic (its singularity lies outside). Green's identity on D minus a small disk of radius ϵ around $\vec{s}_1^{\,\mathrm{proj}}$ gives $\int \nabla\ln\beta_1\cdot\nabla\ln\beta_2 = \oint_{\beta_1=\rho}\ln\beta_1\,\partial_n\ln\beta_2 - \oint_{\beta_1=\epsilon}\ln\beta_1\,\partial_n\ln\beta_2$. On $\beta_1 = \rho$ the factor $\ln\beta_1 = \ln\rho$ is constant and $\oint \partial_n \ln\beta_2\,\mathrm{d}l = \int_D \nabla^2\ln\beta_2 = 0$; the ϵ–circle term is $\mathcal{O}(\epsilon\ln\epsilon) \to 0$. The final bound is $|\vec{u}_2| \leq (r_\perp - \rho)^{-1}$ on D times the disk area. □

This is why the near–region reduction below loses only $\mathcal{O}\big(\eta/\sin^2\theta\big)$ and not the $\mathcal{O}\big(\sqrt{\eta}\big)$ that the term–by–term bound $\int_{\beta_1<\rho}|\vec{u}_1||\vec{u}_2| = 2\pi\rho/r_\perp$ would suggest: the dangerous $\mathcal{O}(1/(\beta_1 r_\perp))$ cross integrand averages to zero azimuthally. For the exact kicks the cancellation is inherited up to the deviations of $\vec{y}_a$ from $\vec{u}_a$, which are bounded in §4 and booked in §6.1.

Lemma 5 (Sequential encounters: a measure estimate)**.** *Call $S \subset \{\beta_1 < \rho\}$ the set of impact parameters whose trajectory, after its close approach to body 1, subsequently passes within ρ of body 2 ("sequential encounters"). At a generic orientation ($\sin\theta \geq \eta^\alpha$) one has, for η small,*

$$\int_S |\vec{y}_2|^2\, \mathrm{d}^2 b \;\leq\; C\,\frac{\eta^2}{\sin^4\theta}\,\ln\frac{1}{\eta}\,, \qquad \int_S |\vec{y}_1||\vec{y}_2|\, \mathrm{d}^2 b \;\leq\; C\,\frac{\eta^2}{\sin^4\theta}\,\ln\frac{1}{\eta}\,. \tag{15}$$

The same bounds hold for the set reaching body 2 before *body 1 (relevant when $\cos\theta < 0$), by time reversal.*

Proof. A trajectory in $\{\beta_1 < \rho\}$ that reaches within s of body 2 must leave the body–1 encounter within an angle $\mathcal{O}(s/r)$ of the direction of body 2, which at a generic orientation lies at an angle $\Theta_{\text{geo}} \geq c\sin\theta$ from the incoming direction (body 2 sits at transverse offset $r_\perp = r\sin\theta$ and longitudinal offset $r\cos\theta$ from body 1). By Lemma 7 below, on $\{\beta_1 < \rho\}$ minus the core the asymptotic–to–osculating map is a near–translation, so the exit direction is the Rutherford function of $\vec{\beta}_1$ up to $\mathcal{O}(\eta)$ relabelings; the measure of osculating impact parameters scattering into a solid–angle patch $\mathrm{d}\Omega$ at angle Θ is $(\mathrm{d}\sigma/\mathrm{d}\Omega)\mathrm{d}\Omega$ with $\mathrm{d}\sigma/\mathrm{d}\Omega = (b_{90,1}/4)^2\sin^{-4}(\Theta/2)$. With $\mathrm{d}\Omega = \mathcal{O}\big(s^2/r^2\big)$ and $\Theta \geq c\sin\theta$,

$$\mu\{\,\vec{b}:\ \text{body–2 pericenter} \in [s,2s]\,\} \;\leq\; C\,\frac{b_{90}^2\, s^2}{r^2\sin^4\theta}. \tag{16}$$

On that set the body–2 kick obeys $|\vec{y}_2| \leq C/s$ (Lemma 1 at the local osculating elements, whose speed is $V[1+\mathcal{O}(\eta/\sin\theta)]$; for $s \lesssim b_{90,2}$ use the uniform cap $|\vec{y}_2| \leq VV_2^{\max}/[G(m_2+m_*)] \leq C/b_{90}$, which follows from $|\Delta\vec{v}_2| \leq \frac{m_*}{m_2+m_*}\,2V_2^{\max}$ and energy conservation). Summing dyadically over $s \in [b_{90}, \rho]$,

$$\int_S |\vec{y}_2|^2 \leq \sum_{\text{dyadic}} \frac{C}{s^2}\cdot\frac{Cb_{90}^2 s^2}{r^2\sin^4\theta} = C\,\frac{\eta^2}{\sin^4\theta}\,\#\{\text{dyadic scales}\} \leq C\,\frac{\eta^2}{\sin^4\theta}\ln\frac{1}{\eta}. \tag{17}$$

For the mixed term use $|\vec{y}_1| \leq C/\max(\beta_1, b_{90}) \leq C/b_{90}$ on the sequential set together with $|\vec{y}_2| \leq C/s$ and the same dyadic count; the product bound is no larger. The deep core $s < b_{90}$ contributes one more dyadic term with the capped kick, within the stated bound. □

Remark 3. *The pointwise statement one might be tempted to make instead — "inside the disk $\beta_1 < \rho$ the other kick $|\vec{y}_2|$ is bounded by $\mathcal{O}(1/r)$" — is* false *for the exact dynamics: a perturber strongly scattered by body 1 can score a direct hit on body 2, with $|\vec{y}_2|$ as large as $\mathcal{O}(1/b_{90})$ (measured violations of the naive bound by factors of several are shown in `three_body_check.py`). It is only the smallness of the measure of such trajectories, Lemma 5, that makes the cross and self–2 contamination of the near disk negligible.*

Lemma 6 (The near–aligned cone)**.** *There is C such that for all orientations (including $\sin\theta < \eta^\alpha$) and η small,*

$$\int_{\Pi_V} |\vec{y}|^2\, \mathrm{d}^2 b \;\leq\; C\ln^3\frac{1}{\eta}. \tag{18}$$

Consequently the cone $\sin\theta < \eta^\alpha$ contributes $\mathcal{O}\big(\eta^{2\alpha}\ln^3(1/\eta)\big)$ to the θ–averages $\int_0^\pi(\cdots)\sin\theta\,\mathrm{d}\theta$ in (39) below.

Proof. By (12) and Cauchy–Schwarz it suffices to bound each $\int |\vec{y}_i|^2$. Split the plane at $\beta_i^{\rm osc} \equiv$ the pericenter distance of the perturber–body–i relative orbit. *(i) Kick bound:* by energy conservation in (1) (all included forces are conservative and the bodies start at rest), $\frac{1}{2}m_* V^2 = \frac{1}{2}m_* v_*^2 + \sum_i \frac{1}{2}m_i v_i^2$ finally, so $|\Delta\vec{v}_i| \le \sqrt{m_*/m_i}\,V$ always; moreover if the trajectory's pericenter w.r.t. body i is p_i, then $|\vec{y}_i| \le C/\max(p_i, b_{90,i})$ (two–body kick at the local osculating elements, Lemma 1, plus the uniform cap above). *(ii) Measure bound:* the set of asymptotic $\vec{b}$ reaching pericenter $\le s$ of body i has measure $\le Cs^2\ln^2(1/\eta)$ for $s \in [b_{90}, r]$. Indeed the angular momentum L_i of the relative orbit about body i changes only by the companion's torque, $|\dot{L}_i| \le |d_i \times g_j|$ with $|g_j| \le Gm_j/q_j^2$; along any incoming path the integrated torque is $\mathcal{O}((Gm_j/V)\ln(1/\eta)) = \mathcal{O}(V b_{90}\ln(1/\eta))$ unless the path first scatters strongly off body j. Since pericenter $\le s$ forces $L_i \lesssim s\,V[1 + 2b_{90}/s]^{1/2} \le CsV$ at pericenter (energy bound on the local speed), the no–strong–scattering part of the set has radius $\le s + C\sqrt{sb_{90}} + Cb_{90}\ln(1/\eta)$, hence measure $\le Cs^2\ln^2(1/\eta)$ for $s \ge b_{90}$. The doubly–scattered part (strong deflection by body j first) is counted as in Lemma 5: trajectories from a dyadic annulus $[p, 2p]$ around body j emerge in an annular cone of opening $\Theta(p) \simeq 2b_{90}/p$ and must hit a disk of radius s at distance $\mathcal{O}(r)$; the feeding measure is $\le Cp^2 \cdot (s^2/r^2)/\Theta(p)^2 = Cs^2p^4/(r^2b_{90}^2) \le Cs^2$ after summing $p \le \rho$. *(iii) Assembly:* dyadically for $s \in [b_{90}, r]$, $\int_{p_i \le r} |\vec{y}_i|^2 \le \sum (C/s^2)(Cs^2\ln^2) \le C\ln^3(1/\eta)$; for $p_i > r$ the straight–line dipole estimate applies ($|\vec{y}| \le Cr/b^2$ once the pericenter bound guarantees $b/2 \le p_i$, which holds for $b \ge Cb_{90}$ by the same angular–momentum argument) and gives a convergent, $\mathcal{O}(1)$ contribution. Finally $\int_0^{\arcsin \eta^\alpha} \sin\theta\,\mathrm{d}\theta = \mathcal{O}(\eta^{2\alpha})$. $\square$

4 The near region: the close encounter is two–body up to controlled errors

Throughout this section the orientation is generic, $\sin\theta \ge \eta^\alpha$, and we work in the near disk of body 1 (body 2 is symmetric). Two scales matter: the distance from the incoming beam to the companion,

$$q \;\ge\; \tfrac{1}{2}\,r_\perp \;=\; \tfrac{1}{2}\,r\sin\theta \tag{19}$$

(the incoming straight line aimed within ρ of body 1 passes body 2 no closer than $r_\perp - \rho \ge r_\perp/2$), and the distance from the *encounter site* to the companion, which is always $r[1 + \mathcal{O}(\sqrt{\eta})]$ (the bodies are r apart). It is the first scale that controls the size of the trajectory relabeling, and the second that controls the local speed.

4.1 Osculating elements and the exact variation–of–parameters identity

Let $\vec{d}(t) = \vec{x}_*(t) - \vec{x}_1(t)$, $\vec{u}(t) = \dot{\vec{d}}(t)$ be the perturber–body–1 relative coordinate. From (1),

$$\ddot{\vec{d}} = -\,\mu_1\frac{\vec{d}}{|\vec{d}|^3} + \vec{g}_2(t), \qquad \mu_1 = G(m_1 + m_*), \qquad \vec{g}_2(t) = -\frac{Gm_2\,(\vec{x}_* - \vec{x}_2)}{|\vec{x}_* - \vec{x}_2|^3}, \tag{20}$$

exactly: body 1 feels only the perturber, so the companion enters the relative dynamics at full monopole strength (not tidally) — this is why the per–trajectory relabeling below is $\mathcal{O}(b_{90})$ and not $\mathcal{O}(b_{90}\,\beta/r)$. Along the incoming leg define the osculating hyperbolic elements of $(\vec{d}, \vec{u})$ with parameter μ_1: energy $E = \frac{1}{2}|\vec{u}|^2 - \mu_1/|\vec{d}|$, speed $V_1 = \sqrt{2E}$, angular momentum $\vec{L} = \vec{d} \times \vec{u}$, Laplace–Runge–Lenz vector $\vec{A} = \vec{u} \times \vec{L} - \mu_1\hat{d}$, eccentricity $e = |\vec{A}|/\mu_1$, incoming asymptote direction $\hat{u}_{\rm in} = (\hat{p} + \sqrt{e^2 - 1}\,\hat{q})/e$ with $\hat{p} = \vec{A}/|\vec{A}|$, $\hat{q} = \hat{L} \times \hat{p}$, and *vector impact parameter* $\vec{\beta}_1 = (\hat{u}_{\rm in} \times \vec{L})/V_1$, $|\vec{\beta}_1| = |\vec{L}|/V_1$ (the order is fixed by the free–motion limit, where $\vec{\beta}_1$ reduces to the transverse offset of the asymptote). Let $t_{\rm p}$ be the time of closest approach to body 1 and write starred quantities for elements evaluated at $t_{\rm p}$.

Since the elements are constants of the unperturbed Kepler flow, they evolve *only* through $\vec g_2$: for any element X,

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \frac{\partial X}{\partial \vec u}\cdot \vec g_2(t), \qquad \text{so} \qquad X^* = X^{\rm asym} + \int_{-\infty}^{t_{\rm p}} \frac{\partial X}{\partial \vec u}\big(\vec d(t), \vec u(t)\big)\cdot \vec g_2(t)\,\mathrm{d}t, \tag{21}$$

an exact identity (variation of parameters), where $X^{\rm asym}$ is the incoming asymptotic value: $V_1^{\rm asym} = V$, $\vec\beta_1^{\rm asym} = \vec b - \vec s_1^{\rm proj}$ (up to the fixed isometry identifying the plane orthogonal to $\hat u_{\rm in}^{\rm asym} = \hat V$ with Π_V), $\hat u_{\rm in}^{\rm asym} = \hat V$. All integrals in (21) converge absolutely: at early times $|\vec d| \simeq V|t|$ and the component of $\partial X/\partial \vec u \cdot \vec g_2$ that could grow (the lever $\propto |\vec d|$ in $\partial\vec\beta_1/\partial\vec u$, $\partial \vec L/\partial \vec u$) multiplies only the *transverse* part of $\vec g_2$, which decays as $1/|t|^3$; the longitudinal part ($\propto 1/t^2$) enters $\dot V_1$ and $\dot\beta_1 \propto \beta_1 \dot V_1/V_1$ with bounded coefficients. (Explicitly: $\partial E/\partial \vec u = \vec u$, $\partial \vec L/\partial \vec u = \vec d\times$, and $\vec\beta_1, \hat u_{\rm in}$ are smooth algebraic functions of $(E, \vec L, \vec A)$ in the hyperbolic regime.)

4.2 The deflection map, now with proofs

Lemma 7 (Incoming–map regularity)**.** *Fix $\lambda \in (0,1]$ and a generic orientation. Let $\mathcal A_\lambda = \{\vec b : \lambda b_{90,1} \le |\vec b - \vec s_1^{\rm proj}| \le \rho\}$. For η small enough the map $\Psi : \vec b \mapsto \vec\beta_1^*$ is a C^1 diffeomorphism of $\mathcal A_\lambda$ onto its image, and, with $K = K(\lambda)$ the Kepler incoming–leg constant of Lemma 11 (Appendix A),*

$$\big|\Psi(\vec b) - (\vec b - \vec s_1^{\rm proj})\big| \;\le\; C\,\frac{b_{90}}{\sin\theta}\,, \tag{22}$$

$$\Big\|\frac{\partial \Psi}{\partial \vec b} - \mathbb{1}\Big\| \;\le\; C\,K\,\frac{\eta}{\sin^2\theta}\,, \qquad \textit{hence} \qquad \Big|\det\frac{\partial \vec b}{\partial \vec\beta_1^*} - 1\Big| \le C\,K\,\frac{\eta}{\sin^2\theta}\,, \tag{23}$$

$$V_1^* \;=\; V\Big[1 + \mathcal O\big(\tfrac{\eta}{\sin\theta}\big)\Big], \qquad b_{90,1}(V_1^*) = b_{90,1}\Big[1 + \mathcal O\big(\tfrac{\eta}{\sin\theta}\big)\Big], \tag{24}$$

$$\hat u_{\rm in}^* \;=\; \hat V + \mathcal O\big(\tfrac{\eta}{\sin\theta}\big)\,. \tag{25}$$

Proof. All statements follow from the exact identity (21) once two ingredients are supplied: bounds on the source, and bounds on the coefficients and their $\vec b$–derivatives.

Source. Along any trajectory of $\mathcal A_\lambda$'s incoming leg, the distance to body 2 satisfies $q_2(t) \ge \max\big(\tfrac12 r_\perp,\ c\,|Vt - r\cos\theta|\big)$ up to $\mathcal O(b_{90})$ corrections (the incoming leg is straight to within accumulated deflections $\mathcal O(b_{90}/\beta)$, and body 2 recoils by only $\mathcal O(b_{90})$ during the relevant span; a standard bootstrap makes this self–consistent). Hence $|\vec g_2(t)| \le Gm_2/q_2(t)^2$ with the displayed floor, and $|\nabla \vec g_2| \le 2Gm_2/q_2^3$.

Speed and tilt. $\dot E = \vec u\cdot\vec g_2$, so $|E^* - \tfrac12 V^2| \le \int |\vec u||\vec g_2|\mathrm{d}t \le CV\cdot\frac{Gm_2}{(r_\perp/2)^2}\cdot\frac{r_\perp}{V}$ + (tails) $= \mathcal O(Gm_2/r_\perp)$, the tails (where $q_2 \simeq V|t|$) contributing $\int^\infty Gm_2/(Vt)^2\,V\mathrm{d}t = \mathcal O(Gm_2/r_\perp)$ as well when cut at $|Vt| \sim r_\perp$. Thus $V_1^{*2} = V^2[1 + \mathcal O\big(Gm_2/(r_\perp V^2)\big)] = V^2[1 + \mathcal O(\eta/\sin\theta)]$, which is (24); note the osculating energy is shifted by the *work* done by the companion along the incoming leg (positive when the leg descends the companion's potential — the infall speeds the encounter up). The same estimate applied to $\vec A/\mu_1$ and $\hat u_{\rm in}$ gives (25).

Displacement. For $X = \vec\beta_1$, split the integrand of (21) into the transverse and longitudinal parts of $\vec g_2$ relative to $\hat V$. The coefficient $|\partial\vec\beta_1/\partial\vec u| \le C(|\vec d|/V_1 + \beta_1/V_1)$ (explicit differentiation of the element formulas; Appendix A). With $|\vec d(t)| \simeq \max(\beta_1, V|t - t_{\rm p}|)$: near the companion's longitudinal position ($Vt \simeq r\cos\theta$, duration $\sim q/V$) the transverse source is $\le Gm_2/q^2$, the lever is $\le Cr/V$, giving $C(r/V)(Gm_2/q^2)(q/V) = C\,Gm_2\,r/(qV^2) \le Cb_{90}/\sin\theta$; away from it the transverse part of $\vec g_2$ decays one power of distance faster than $1/q_2^2$ and the integral is dominated by the same scale; the longitudinal part enters through $\dot V_1$ as above, contributing $\mathcal{O}(\beta_1 \cdot \eta/\sin\theta) \le \mathcal{O}\big(b_{90}\sqrt{\eta}/\sin\theta\big)$. This is (22). (At $\cos\theta = 0$ the "lever" formulation degenerates but the same integral gives $\mathcal{O}\big(Gm_2/V^2\big) = \mathcal{O}(b_{90})$ directly.)

Derivative. Differentiate (21) w.r.t. $\vec b$:

$$\frac{\partial\vec\delta}{\partial\vec b} = \int_{-\infty}^{t_\mathrm{p}} \Bigg[\underbrace{\Big(\nabla_{(\vec d,\vec u)}\frac{\partial\vec\beta_1}{\partial\vec u}\Big)\cdot\frac{\partial(\vec d,\vec u)}{\partial\vec b}\cdot\vec g_2}_{\text{(I)}} + \underbrace{\frac{\partial\vec\beta_1}{\partial\vec u}\cdot\nabla\vec g_2\cdot\frac{\partial\vec x_*}{\partial\vec b}}_{\text{(II)}}\Bigg]\mathrm{d}t + (\text{boundary term at } t_\mathrm{p}), \tag{26}$$

where $\vec\delta \equiv \Psi(\vec b) - (\vec b - \vec s_1^{\,\mathrm{proj}})$. By Lemma 11 the incoming–leg tangent flow of the reference Kepler problem is bounded, $\|\partial(\vec d,\vec u)(t)/\partial(\text{asymptotic data})\| \le K(\lambda)$ in the natural scalings, uniformly on $\beta_1 \ge \lambda b_{90,1}$ and in time; a Gronwall estimate with the perturbation $\int \|\nabla\vec g_2\| K\,\mathrm{d}t = \mathcal{O}\big(K\eta/\sin^2\theta\big) \ll 1$ transfers the same bound (with doubled constant) to the true flow. Term (II) is then bounded by $\int C(|\vec d|/V)\cdot(2Gm_2/q_2^3)\cdot K\,\mathrm{d}t \le CK\,Gm_2 r/(q^2V^2)\cdot(1/q)\cdot q = CK\,Gm_2\,r/(q^2V^2) \le CK\,\eta/\sin^2\theta$ by the same time–slicing as above (one extra power of q in the denominator relative to the displacement estimate, one extra factor K from the tangent flow). Term (I) carries the derivative of the algebraic coefficient, again bounded by K times $1/(\text{local scale})$, and gives the same order. The boundary term is the variation of t_p itself, which enters only through elements evaluated at pericenter — but the elements are stationary under the Kepler part of the flow, so this term is proportional to $\vec g_2(t_\mathrm{p}) = \mathcal{O}\big(Gm_2/r^2\big)$ times $\partial t_\mathrm{p}/\partial\vec b = \mathcal{O}(K/V)$, i.e. $\mathcal{O}(K\eta\, b_{90}/r)$, negligible. Collecting, $\|\partial\vec\delta/\partial\vec b\| \le CK\eta/\sin^2\theta$, which is (23); for η small the map is a local diffeomorphism, and injectivity on $\mathcal{A}_\lambda$ follows since Ψ is a perturbation of the identity by $\vec\delta$ with small Lipschitz constant. $\square$

Remark 4 (What the map does and does not say)**.** *The displacement* (22) *is* $\mathcal{O}(b_{90})$ *and does* not *vanish relative to the cutoff: the companion's monopole shifts each individual trajectory's impact parameter by an* $\mathcal{O}(1)$ *fraction of* b_{90} *— this is the* $\sim$ 25% *per–trajectory discrepancy seen in the numerics, independent of* r*. What makes it harmless is* (23)*: the shift is a near–*translation *of the impact–parameter plane, so it relabels trajectories without changing the measure, to* $\mathcal{O}\big(\eta/\sin^2\theta\big)$*. The two statements are logically independent, and both are verified separately in* ***deflection_map_check.py****: the vector map must be used, since the* $\mathcal{O}(1)$ *translation also rotates* $\vec\beta_1$ *by an* $\mathcal{O}(b_{90}/\beta_1)$ *angle that a scalar diagnostic would misread as a Jacobian distortion.*

Lemma 8 (Kick fidelity)**.** *In the setting of Lemma 7, for* $\vec b \in \mathcal{A}_\lambda \setminus S$ *(the sequential set* S *of Lemma 5 removed),*

$$\vec y_1(\vec b) \;=\; \vec y_1^{\,2\mathrm{B}}\big(\vec\beta_1^{\,*}, V_1^*\big)\,\big[1+\epsilon_1(\vec b)\big] + \vec\Delta(\vec b), \qquad |\epsilon_1| \;\le\; C\,K(\lambda)\,\frac{\eta}{\sin^2\theta}\,\ln\frac{1}{\eta}\,, \qquad |\vec\Delta| \;\le\; \frac{C}{r\sin\theta}\,, \tag{27}$$

where $\vec y_1^{\,2\mathrm{B}}(\vec\beta_1, V_1)$ *is the exact isolated two–body kick of Lemma 1 evaluated at the osculating pericenter elements, with magnitude*

$$\big|\vec y_1^{\,2\mathrm{B}}\big|^2 = \Big(\frac{V}{V_1^*}\Big)^2\frac{1}{\beta_1^{*2} + b_{90,1}(V_1^*)^2}\,, \qquad b_{90,1}(V_1^*) = \frac{G(m_1+m_*)}{V_1^{*2}}. \tag{28}$$

The additive piece $\vec\Delta$ *is the companion's direct impulse on the perturber routed through the relative–coordinate bookkeeping; it is bounded, nearly* $\vec b$*–independent across the near disk (it varies on the scale* $r\sin\theta$*), and contributes* $\mathcal{O}\big(\sqrt{\eta}/\sin\theta\big)$ *to the near–region integrals (see below), on top of which its azimuthal average against* $\vec y_1^{\,2\mathrm{B}}$ *nearly cancels.*

Note the prefactor $(V/V_1^*)^2 = 1-\mathcal{O}(\eta/\sin\theta)$ in (28): the kick normalization $\vec{y}_1 = (V/2Gm_*)\Delta\vec{v}_1$ carries the *asymptotic* speed while the encounter runs at the boosted local speed. This factor multiplies the whole logarithm and is of the same order as the Jacobian correction (with the opposite sign); both sit inside the error budget of the theorem, but neither may be dropped silently in (27)–(28).

Proof. Body 1 feels only the perturber, so exactly $\Delta\vec{v}_1 = Gm_*\int \hat{d}/d^2\,\mathrm{d}t$ along the true relative orbit. From (20), $\int \mu_1\hat{d}/d^2\,\mathrm{d}t = -\Delta\vec{u} + \int \vec{g}_2\,\mathrm{d}t$, whence

$$\Delta\vec{v}_1 = \frac{m_*}{m_1+m_*}\Big[-\Delta\vec{u} + \int \vec{g}_2\,\mathrm{d}t\Big], \qquad \Delta\vec{v}_1^{\,2\mathrm{B}} = -\frac{m_*}{m_1+m_*}\,\Delta\vec{u}^{\,0}, \tag{29}$$

where $\vec{d}^{\,0}(t)$ is the isolated Kepler orbit (parameter μ_1) with the same position and velocity as the true relative orbit at t_p; its asymptotic elements are exactly $(\vec{\beta}_1^*, V_1^*)$ and its body–1 kick has magnitude (28) by Lemma 1 (the factor $(V/V_1^*)^2$ arising because $\vec{y}_1$ is normalized with the asymptotic V). The additive term is $\vec{\Delta} = (V/2Gm_*)\frac{m_*}{m_1+m_*}\int \vec{g}_2\,\mathrm{d}t$: since the trajectory passes the companion no closer than $q \ge r_\perp/2$, $|\int \vec{g}_2\,\mathrm{d}t| \le CGm_2/(qV)$, giving $|\vec{\Delta}| \le C/(r\sin\theta)$, essentially constant across the near disk.

For the multiplicative error, set $\vec{\xi} = \vec{d} - \vec{d}^{\,0}$, $\vec{\xi}(t_\mathrm{p}) = \dot{\vec{\xi}}(t_\mathrm{p}) = 0$, $\ddot{\vec{\xi}} = \nabla F_\mathrm{Kep}(\vec{d}^{\,0})\cdot\vec{\xi} + \mathcal{O}\big(|\vec{\xi}|^2\big) + \vec{g}_2$. By Lemma 11 the Kepler transition matrices from t_p are bounded by $K(\lambda)$ in scaled units on both legs, so Duhamel and Gronwall give $|\vec{\xi}(t)| \le CK\,(Gm_2/q_2^2)\,(t-t_\mathrm{p})^2$ while $|\vec{\xi}| \ll |\vec{d}^{\,0}|$. Then, with $d(t) \simeq \max(\beta_1^*, V_1|t-t_\mathrm{p}|)$ and $|\Delta\vec{u}^{\,0}| \ge c(\lambda)\,\mu_1/(V_1\beta_1^*)$,

$$\frac{|\Delta\vec{u}-\Delta\vec{u}^{\,0}|}{|\Delta\vec{u}^{\,0}|} \;\le\; C\,\frac{V_1\beta_1^*}{\mu_1}\cdot\mu_1\!\int \frac{2\,|\vec{\xi}(t)|}{\min(d,d^0)^3}\,\mathrm{d}t \;\le\; CK\,\frac{Gm_2\,\beta_1^*}{q^2V^2}\,\ln\frac{r}{\beta_1^*} \;\le\; CK\,\frac{\eta^{3/2}}{\sin^2\theta}\,\ln\frac{1}{\eta}\,, \tag{30}$$

the middle bound collecting the near zone $|t-t_\mathrm{p}| \le \rho/V$ (where the t^2–growth of $\vec{\xi}$ meets the $1/d^3$ decay of the tidal kernel in a logarithm) and the far legs (where the kernel decays and $\vec{\xi}$ grows at most quadratically in d until the companion is passed); the last step uses $\beta_1^* \le \rho$, i.e. $\beta_1^*/r \le \sqrt{\eta}$, and $q \ge r_\perp/2$. This is stronger than the stated $|\epsilon_1|$ bound. On the sequential set the far–leg estimate fails (the true outgoing leg is re–scattered by body 2), which is why S is excluded here and handled by measure in Lemma 5. All estimates bound *vector* differences, so they apply to the tensor $\vec{y}_1\otimes\vec{y}_1$ as well.

Finally, the integrated effect of $\vec{\Delta}$ on the near–region self–energy: the square term is $\int_{\beta_1<\rho}|\vec{\Delta}|^2 \le C\rho^2/(r\sin\theta)^2 = \mathcal{O}\big(\eta/\sin^2\theta\big)$, and the cross term is bounded crudely by

$$\int_{\beta_1<\rho} 2\,|\vec{y}_1^{\,2\mathrm{B}}|\,|\vec{\Delta}|\,\mathrm{d}^2b \;\le\; \frac{C}{r\sin\theta}\int_0^\rho \frac{2\pi\beta\,\mathrm{d}\beta}{\max(\beta,b_{90})} \;=\; \mathcal{O}(\sqrt{\eta}/\sin\theta)$$

— with an extra azimuthal cancellation (the direction of $\vec{y}_1^{\,2\mathrm{B}}$ sweeps the circle while $\vec{\Delta}$ is fixed) that we do not need. □

4.3 Near–region self–energy

Fix a generic orientation and let $\lambda = \lambda(\eta) \to 0$ slowly (specified below). Split $\{\beta_1^\mathrm{asym} < \rho\}$ into the annulus $\mathcal{A}_\lambda$ and the core. On the annulus, change variables $\vec{b} \to \vec{\beta}_1^*$ (Lemma 7) and use Lemma 8:

$$\begin{aligned}\int_{\mathcal{A}_\lambda} |\vec{y}_1|^2\,\mathrm{d}^2b &= \int_{\lambda b_{90,1}\lesssim\beta_1\lesssim\rho} \Big(\frac{V}{V_1^*}\Big)^2 \frac{1+\mathcal{O}(\epsilon_1)}{\beta_1^2 + b_{90,1}(V_1^*)^2}\,\Big|\det\tfrac{\partial\vec{b}}{\partial\vec{\beta}_1}\Big|\,\mathrm{d}^2\beta_1 \;+\; (\vec{\Delta}\text{ terms}) \\ &= 2\pi\ln\frac{\rho}{b_{90,1}} - \pi\ln(1+\lambda^2) + \mathcal{E}_\mathrm{ann},\end{aligned} \tag{31}$$

with, by (23)–(28) and (27), $|\mathcal{E}_{\rm ann}| \le C\,K(\lambda)\,\frac{\eta}{\sin^2\theta}\ln^2\frac{1}{\eta}$ (the relative errors — Jacobian, speed prefactor, cutoff shift, ϵ_1 — multiply a logarithm) + $\mathcal{O}\big(\sqrt{\eta}/\sin\theta\big)$ (two contributions of the same size: the $\mathcal{O}(b_{90}/\sin\theta)$ mismatch between the asymptotic–$\vec b$ annulus boundaries and the osculating–β_1 ones, an annulus of relative width $b_{90}/(\rho\sin\theta)$ integrated against $\mathrm{d}^2\beta_1/\beta_1^2$; and the additive $\vec\Delta$ cross term of Lemma 8).

The core needs *no* fidelity estimate: by the energy cap of Lemma 6(i), $|\vec y_1| \le C/b_{90,1}$ there, while the preimage measure of the core disk is $\pi\lambda^2 b_{90,1}^2[1+\mathcal{O}\big(K\eta/\sin^2\theta\big)]$ (its boundary is the diffeomorphic image of the circle $\beta_1^* = \lambda b_{90,1}$, and the enclosed area follows from Stokes applied to the boundary parametrization; no extra preimage components exist at generic orientation because, outside S, the map is a global near–translation on the disk). Hence the true core contribution and the two–body core value $\pi\ln(1+\lambda^2) \le \pi\lambda^2$ are *each* $\mathcal{O}\big(\lambda^2\big)$, so replacing one by the other costs $\mathcal{O}\big(\lambda^2\big)$. Adding the annulus,

$$\boxed{\int_{\beta_1<\rho} |\vec y_1|^2\,\mathrm{d}^2b = 2\pi\ln\frac{\rho}{b_{90,1}} \;+\; \mathcal{O}\big(\lambda^2\big) \;+\; \mathcal{O}\Big(K(\lambda)\,\frac{\eta}{\sin^2\theta}\ln^2\frac{1}{\eta}\Big) \;+\; \mathcal{O}\Big(\frac{\sqrt{\eta}}{\sin\theta}\Big)} \tag{32}$$

and likewise about body 2. With $K(\lambda) \le C\lambda^{-p}$ (Lemma 11), the choice $\lambda = \eta^{(1-2\alpha)/(p+2)}$ balances the first two error terms on $\sin\theta \ge \eta^\alpha$ and makes the whole error $\mathcal{O}\big(\eta^{c_0}\ln^2(1/\eta)\big)$ with $c_0 = \min\big\{\frac{2(1-2\alpha)}{p+2}, \frac{1}{2}-\alpha\big\} > 0$; the literal $b_{90,1}$ appears with unit coefficient and *no additive constant survives*. Finally, the near disk's contribution to the full $|\vec y|^2$ differs from (32) by the cross and self–2 pieces, bounded by Lemma 4 (plus the deviations $\vec y_a - \vec u_a$, which contribute $\mathcal{O}\big(\eta\ln(1/\eta)/\sin^2\theta\big)$ by the same fidelity estimates) and by Lemma 5: all within the error already displayed. By Lemma 3 and the same error budget, the near disks contribute $0 + \mathcal{O}(\text{same})$ to the anisotropy $\mathcal{Y}_\perp - \mathcal{Y}_\parallel$.

5 The far region: straight lines, the upper scale, and the anisotropy

In the far region every impact parameter exceeds $\rho \gg b_{90}$, so the trajectory is a small perturbation of a straight line.

Lemma 9 (Far–region kick). *On $\{\beta_1, \beta_2 \ge \rho\}$ (generic orientation) the relative kick equals its straight–line value*

$$\vec y^{(0)} = \nabla_b\big[\ln|\vec b - \vec s_2^{\,\rm proj}| - \ln|\vec b - \vec s_1^{\,\rm proj}|\big] = \frac{\vec b - \vec s_2^{\,\rm proj}}{|\vec b - \vec s_2^{\,\rm proj}|^2} - \frac{\vec b - \vec s_1^{\,\rm proj}}{|\vec b - \vec s_1^{\,\rm proj}|^2} \tag{33}$$

(the overall sign is opposite to $\vec y$ as defined from $\Delta\vec v_1 - \Delta\vec v_2$; irrelevant for second moments) up to a relative error $\le c\,b_{90}/\beta_{\min} \le c\sqrt{\eta}$, where $\beta_{\min} = \min(\beta_1,\beta_2)$; for $b \gg r$ the error of the dipole *is relative $\mathcal{O}(b_{90}/r)$, since the $\mathcal{O}(b_{90})$ displacement of the trajectory shifts each monopole term coherently. Consequently*

$$\int_{\rm far} (y^{(0)})^2 \text{ is convergent at large } b \text{ (dipole cancellation)}; \tag{34}$$

$$\int_{\Pi_V\setminus(D_1\cup D_2)} \big[(\vec y^{(0)})_\perp^2 - (\vec y^{(0)})_\parallel^2\big]\,\mathrm{d}^2b = 2\pi + \mathcal{O}\big(\rho^2/r_\perp^2\big) \quad (\textit{anisotropy}), \tag{35}$$

$$\int_{\beta_i\ge\rho} (\vec y^{(0)})^2\,\mathrm{d}^2b = 2\pi\ln\frac{r_\perp^2}{\rho^2} + \mathcal{O}\big(\rho^2/r_\perp^2\big) \quad (\textit{trace, no constant}), \tag{36}$$

where $D_a = \{\beta_a < \rho\}$ are the circular near disks.

Proof. Error of the straight line. A trajectory with $\beta_{\min} \geq \rho$ is deflected by total angle $\leq 2b_{90}/\beta_{\min}$ (sum of the two two–body deflections, each $\leq b_{90,i}/\beta_i$), so its transverse excursion from the straight line is $\leq c\, b_{90}$ over the relevant arc and the kick differs from (33) by the stated relative $\mathcal{O}(b_{90}/\beta_{\min})$; at $b \gg r$ the excursion shifts both monopole terms by the same $\mathcal{O}(b_{90})$, so the dipole error is controlled by the field *gradient*, one factor b_{90}/b better relative to each monopole, i.e. relative $\mathcal{O}(b_{90}/r)$ on the dipole.

Convergence. For $b \gg r$, $\vec{y}^{(0)}$ is the gradient of a dipole, $|\vec{y}^{(0)}| = \mathcal{O}\big(r/b^2\big)$, so $\int (\vec{y}^{(0)})^2\, b\, \mathrm{d}b = \mathcal{O}\big(\int r^2 b^{-3} \mathrm{d}b\big) < \infty$: no outer cutoff is needed.

Anisotropy. With $w = b_{(1)} + i b_{(2)}$ and the two bodies at real coordinates s_1, s_2 ($s_2 - s_1 = r_\perp$), the vector $\vec{u}_a$ corresponds to $(\bar{w} - s_a)^{-1}$ and $(y_\perp^{(0)})^2 - (y_\parallel^{(0)})^2 = \mathrm{Re}\,(Y^2)$ with $Y = (\bar{w} - s_2)^{-1} - (\bar{w} - s_1)^{-1}$. Partial fractions give

$$Y^2 = \frac{1}{(\bar{w} - s_1)^2} + \frac{1}{(\bar{w} - s_2)^2} - \frac{2}{r_\perp}\Big[\frac{1}{\bar{w} - s_2} - \frac{1}{\bar{w} - s_1}\Big]. \tag{37}$$

The last bracket is absolutely integrable near each body, and by the disk identity $\int_{|w|<\Lambda} (\bar{w} - s)^{-1} \mathrm{d}^2 b = -\pi s$ (uniform–disk Cauchy transform; Λ–independent for $|s| < \Lambda$) its full–plane integral is $-\pi(s_2 - s_1) = -\pi r_\perp$; the excised disks change this by $\mathcal{O}(\rho \cdot \rho/r_\perp)$. The squared self terms require care: they are only *conditionally* convergent at their singularities, and their value depends on the shape of the excision. Under excision of a *circular* disk — of any center containing the singularity — they vanish identically: $\int_{|w-c|<R,\,|w-s|>\epsilon} (\bar{w} - s)^{-2} \mathrm{d}^2 b = 0$, by the residue computation $\oint_{|w-c|=R} \frac{\mathrm{d}\bar{w}}{w-s} = 0$ (the poles at $w = s$ and the double pole from $\bar{w} = \bar{c} + R^2/(w-c)$ cancel) together with the vanishing of the ϵ–circle term. This shape–dependence is not academic: excising an *ellipse* of axis ratio a/b instead leaves $-\pi(a-b)/(a+b)$ per self term ($-\pi/3$ at $a/b = 2$, i.e. 17% of the entire 2π), as one checks by the same contour computation. The decomposition of §3 uses circular disks, and the deflection map distorts their images only at $\mathcal{O}(\eta)$, so the self terms drop and $\int \mathrm{Re}(Y^2) = (-2/r_\perp)(-\pi r_\perp) = 2\pi$ up to the stated error, which is (35).

Trace. Writing $\vec{u}_a = (\vec{b} - \vec{s}_a^{\,\mathrm{proj}})/\beta_a^2$ and introducing an auxiliary outer cutoff Λ, the two self terms are $\int_{\rho \leq \beta_a,\, |w|<\Lambda} u_a^2 = 2\pi \ln(\Lambda/\rho) + \mathcal{O}\big(\rho^2/r_\perp^2\big) + \mathcal{O}(r/\Lambda)$ each (the second error from the off–centred outer boundary, vanishing as $\Lambda \to \infty$; the first from excising the *other* body's disk), and the cross term over the full disk is $\int_{|w|<\Lambda} \vec{u}_1 \cdot \vec{u}_2 = 2\pi \ln(\Lambda/r_\perp)$ (via $\nabla^2 \ln \beta_2 = 2\pi \delta^{(2)}$ and Green's theorem), while its restriction to each near disk vanishes *exactly* by Lemma 4 — this exact vanishing is what keeps the trace bookkeeping at $\mathcal{O}\big(\rho^2/r_\perp^2\big)$ rather than the $\mathcal{O}\big(\sqrt{\eta}\big)$ a term–by–term bound would give. Hence

$$\int_{\beta_i \geq \rho} (\vec{y}^{(0)})^2 = \underbrace{2\,(2\pi) \ln \tfrac{\Lambda}{\rho}}_{\text{self}} - \underbrace{2\,(2\pi) \ln \tfrac{\Lambda}{r_\perp}}_{\text{cross}} + \mathcal{O}\big(\rho^2/r_\perp^2\big) = 2\pi \ln \frac{r_\perp^2}{\rho^2} + \mathcal{O}\big(\rho^2/r_\perp^2\big)\,,$$

the auxiliary Λ cancelling and *no* additive constant remaining, which is (36). The constant 2π lives in the anisotropy, never in the trace. □

The far region thus supplies (a) the upper scale $r_\perp$ of the logarithm and (b) the entire anisotropy $\mathcal{Y}_\perp - \mathcal{Y}_\parallel = 2\pi$, both cutoff–independent, with straight–line accuracy $\mathcal{O}\big(\sqrt{\eta}\big)$ near the matching radius and $\mathcal{O}(\eta)$ in the dipole bulk.

6 Assembly: proof of Theorem 1

6.1 The in–plane tensor and the error tally

Fix a generic orientation. Add the two near results (32) and the far trace (36); the isotropic (trace) part is

$$\mathcal{Y}_\perp + \mathcal{Y}_\parallel = \underbrace{2\pi \ln \frac{\rho}{b_{90,1}} + 2\pi \ln \frac{\rho}{b_{90,2}}}_{\text{near 1,2}} + \underbrace{2\pi \ln \frac{r_\perp^2}{\rho^2}}_{\text{far}} + \big[\text{errors}\big] = 2\pi \ln \frac{r_\perp^2}{b_{90,1} b_{90,2}} + \big[\text{errors}\big]. \tag{38}$$

The intermediate scale ρ cancels identically, leaving the literal $b_{90,1}b_{90,2}$ and the upper scale $r_\perp^2$, with *no additive constant.* The error tally, at orientation θ with $\sin\theta \geq \eta^\alpha$ and with $\lambda = \eta^{(1-2\alpha)/(p+2)}$:

core replacement (twice)	$\mathcal{O}\big(\lambda^2\big) = \mathcal{O}\big(\eta^{2(1-2\alpha)/(p+2)}\big)$
map Jacobian, speed boost $(V/V_1)^2$, $b_{90}(V_1)$, kick fidelity	$\mathcal{O}\big(K(\lambda)\,\eta \ln^2(1/\eta)/\sin^2\theta\big)$
annulus/disk boundary mismatch	$\mathcal{O}\big(\sqrt{\eta}/\sin\theta\big)$
cross and self–2 contamination of the near disks	$\mathcal{O}\big(\eta \ln(1/\eta)/\sin^2\theta\big)$ (Lemmas 4, 5)
far–region straight–line error	$\mathcal{O}\big(\sqrt{\eta}\big)$
far–region excised–disk terms	$\mathcal{O}\big(\rho^2/r_\perp^2\big) = \mathcal{O}\big(\eta/\sin^2\theta\big)$

With $K(\lambda) \leq C\lambda^{-p}$ every entry is $\mathcal{O}(\eta^c)$ for some $c = c(\alpha, p) > 0$, uniformly on $\sin\theta \geq \eta^\alpha$; this proves the first display of (6). For the anisotropy, the near regions contribute 0 up to the same tally (Lemma 3 and §4.3) and the far region gives $\mathcal{Y}_\perp - \mathcal{Y}_\parallel = 2\pi + \mathcal{O}\big(\sqrt{\eta}\big)$ (Lemma 9); this proves the second display. *Numerically* the whole correction is $\mathcal{O}(\eta)$ (§7); the weaker proven rate is all the constants require.

6.2 Angular and speed averages give the constants

With $\mathcal{Y}_\perp = \pi[\bar{L}+1]$, $\mathcal{Y}_\parallel = \pi[\bar{L}-1]$, $\bar{L} = \ln[r_\perp^2/(b_{90,1}b_{90,2})]$ and $r_\perp = r\sin\theta$, the projections of the companion paper give (Lemma 3 ensuring the longitudinal recoil is carried correctly by the trace)

$$Q_r = 2C_Q \int_0^\pi \mathcal{Y}_\perp \sin^3\theta \,\mathrm{d}\theta, \qquad Q_r + 2Q_t = 2C_Q \int_0^\pi (\mathcal{Y}_\perp + \mathcal{Y}_\parallel) \sin\theta\,\mathrm{d}\theta, \tag{39}$$

which are *elementary and exact*: using $\int_0^\pi \sin^3\theta\,\mathrm{d}\theta = \frac{4}{3}$, $\int_0^\pi \sin^3\theta \ln\sin\theta\,\mathrm{d}\theta = \frac{4}{3}\ln 2 - \frac{10}{9}$, $\int_0^\pi \sin\theta\,\mathrm{d}\theta = 2$, $\int_0^\pi \sin\theta\ln\sin\theta\,\mathrm{d}\theta = 2\ln 2 - 2$, and the Maxwellian identity $\langle V^{-1}\ln V^4\rangle = \big[\ln(4\sigma^4) - 2\gamma_{\mathrm{E}}\big]\langle V^{-1}\rangle$ for the V–dependence of $b_{90,i} \propto V^{-2}$, one obtains exactly (5) with $\mathcal{L} = \ln[16\sigma^4 r^2/(G^2(m_1 + m_*)(m_2+m_*))] - 2\gamma_{\mathrm{E}}$. The error terms enter these averages linearly: the generic–orientation errors integrate to $\mathcal{O}(\eta^c \ln(1/\eta))$ (each $1/\sin^k\theta$ weight is tamed by the $\sin\theta$ or $\sin^3\theta$ measure and the cone exclusion, e.g. $\int_{\sin\theta\geq\eta^\alpha} \frac{\eta\ln^2(1/\eta)}{\sin^2\theta}\sin\theta\,\mathrm{d}\theta = \mathcal{O}\big(\alpha\,\eta\ln^3(1/\eta)\big)$), and the cone contributes $\mathcal{O}\big(\eta^{2\alpha}\ln^3(1/\eta)\big)$ by Lemma 6. It remains to control the speed average.

Lemma 10 (Low–speed tail)**.** *For a Maxwellian bath, the contribution to Q_r, Q_t of speeds $V \leq V_c$, for any $V_c \leq \sigma$, is bounded by $C\,G^2 m_*^2 n\sigma^{-1}\cdot\big[(m+m_*)^2/m_*^2\big]\,(V_c/\sigma)^2$, and the closed–form integrand is bounded by the same quantity there. In particular, choosing $V_c^2 = G(m+m_*)/r$, the tail where the impulsive parameter $\eta(V) = G(m+m_*)/(rV^2)$ is* not *small contributes a relative $\mathcal{O}\big(Gm/(r\sigma^2)\big)$ to (5), while for $V > V_c$ the fixed–V errors of §6.1, $\mathcal{O}(\eta(V)^c\,\text{polylog})$, average to $\mathcal{O}\Big(\big(Gm/(r\sigma^2)\big)^{c'}\Big)$ for some $c' > 0$. Hence (7) holds in the joint limit $Gm/(r\sigma^2) \to 0$.*

Proof. Energy conservation gives the uniform cap $|\Delta\vec{v}| \le CV$ (proof of Lemma 6(i)). For $b \ge B \equiv C' b_{90}(V) + 2r$ the angular–momentum argument of Lemma 6(ii) keeps the pericenter above $b/2$, and the tidal–dipole estimate gives $|\Delta\vec{v}| \le CGm_* r/(Vb^2)$, so $\int_{b>B} |\Delta v|^2 \mathrm{d}^2 b \le CG^2 m_*^2 r^2/(V^2 B^2)$; for $b < B$ use the cap: $\int_{b<B} \le CV^2 B^2$. Hence, for $V \le V_c$,

$$\int |\Delta v|^2 \,\mathrm{d}^2 b \le C\Big[V^2 b_{90}(V)^2 + V^2 r^2 + \frac{G^2 m_*^2 r^2}{V^2 b_{90}(V)^2}\Big] \le C\,\frac{G^2(m+m_*)^2}{V^2}$$

(all three terms reduce to this using $b_{90}(V) = G(m+m_*)/V^2 \ge r$ there). The V–integral of the tail is then $n\int_0^{V_c} 4\pi V^2 g\, V \cdot CG^2(m+m_*)^2 V^{-2}\mathrm{d}V = \mathcal{O}\big(G^2(m+m_*)^2\, n\, V_c^2/\sigma^3\big)$, which is the stated bound; the comparison with the main term $\mathcal{O}\big(G^2 m_*^2 n \sigma^{-1} \ln\big)$ gives the relative $\mathcal{O}\big(Gm/(r\sigma^2)\big)$ upon inserting $V_c^2 = G(m+m_*)/r$. For $V > V_c$ the error weight $\langle V^{-1}\eta(V)^c\rangle/\langle V^{-1}\rangle = \mathcal{O}\Big(\eta(\sigma)^{c'}\Big)$ since $\int_0^\infty V e^{-V^2/2\sigma^2} V^{-2c}\mathrm{d}V$ converges for $c < 1$. □

Because the in–plane tensor is correct up to the tally of §6.1, and the averages are linear in it, the constants $-\frac{2}{3}, -\frac{8}{3}$ emerge with a vanishing correction; that is (7). ■

6.3 Why no constant can hide

The proof pins the constant three times over: (i) Lemma 2 fixes the close–encounter scale to the bare $b_{90,i}$ with zero additive constant (the longitudinal recoil cancels the soft–core $\sqrt{e}$); (ii) the intermediate scale ρ cancels in (38), so no scheme scale intrudes; (iii) the anisotropy 2π is the value of a convergent integral, independent of any cutoff — provided the excisions are circular, which the decomposition guarantees and the deflection map preserves to $\mathcal{O}(\eta)$. A different multiplicative choice $b_{90,i} \to \alpha b_{90,i}$ would shift $\mathcal{Y}_\perp + \mathcal{Y}_\parallel$ by the *constant* $-2\pi\ln(\alpha^2)$ and hence $Q_{r,t}$ by $-\frac{8\pi}{3} C_Q \ln\alpha^2 \ne 0$; the proof leaves no room for such a shift, and the numerics of §7 exclude it directly.

7 The surviving correction: size, sign, and interpretation

Theorem 1 establishes the limit; we now discuss the leading correction, more carefully than a single catch–phrase allows.

7.1 What survives at $\mathcal{O}(\eta)$

Three effects of the companion survive at relative order η on the near–region logarithms, and they do not all have the same sign:

- the *measure* (Jacobian) distortion of the impact–parameter relabeling, (23) — the gravitational–focusing–like compression of the beam, which *increases* the close–encounter rate;
- the *speed boost* $(V/V_1^*)^2 = 1-\mathcal{O}(\eta)$ of (28), which multiplies the whole logarithm and *decreases* it (the encounter runs faster than the asymptotic normalization assumes);
- the shrinking of the local cutoff, $b_{90,1}(V_1^*) < b_{90,1}$, an *additive* $+\mathcal{O}(\eta)$ on the log.

The proven statement is only that their sum is within the error tally of §6.1; we do *not* claim the net correction equals the textbook focusing factor $1 + 2Gm_2/(rV^2)$ — the areal compression of a cold beam at a point is geometry–dependent, and the three effects partially cancel. What the numerics below establish is that the *net* correction is a clean $\mathcal{O}(\eta)$: positive in the configurations tested, of magnitude $\approx 1.7\,\eta$ at $m_1 = m_2 = m_*$ (where $2Gm_2/(rV^2) = \eta$), with no constant offset. Since an $\mathcal{O}(\eta)$ correction multiplying $\ln(1/\eta)$ still vanishes, the constants are untouched in the limit, as Theorem 1 states. Per trajectory, by contrast, the relabeling (22) is $\mathcal{O}(b_{90})$ — an $\mathcal{O}(1)$ fraction of the cutoff — and never becomes small; it is its area–preserving character (23) that erases it from the integral.

7.2 Direct three–body tests

Two scripts confront the proof with the exact dynamics (1).

`three_body_check.py` integrates the genuine encounter (perturber in the field of both recoiling bodies, binary internal force off), accumulates $\Delta\vec{v}_1, \Delta\vec{v}_2$, and forms the binary $\mathcal{Q}^{\alpha\beta}$ by Monte Carlo, comparing to the superposition ansatz by common random numbers. The integrator reproduces the exact two–body kick (8) in the $m_2 \to 0$ limit to $\sim 10^{-3}$. Findings: (i) per trajectory the deflection is $\mathcal{O}(1)$ and does not vanish ($\sim 25\%$ at $\beta_1 \simeq b_{90}$, independent of r, proportional to m_2 — the monopole relabeling (22)); (ii) in the integral it is $\mathcal{O}(\eta)$: holding V fixed and shrinking η, the trace enhancement $[\mathcal{Q}^{\alpha}{}_{\alpha}]_{\rm 3body}/[\mathcal{Q}^{\alpha}{}_{\alpha}]_{\rm sup} - 1$ is

η	0.10	0.05	0.025
enhancement	+0.173	+0.090	+0.042

halving as η halves, positive, extrapolating to zero with no constant offset. A constant cutoff shift $b_{90} \to \alpha b_{90}$ would instead drive this ratio to the nonzero $-\ln\alpha^2$; it does not.

`deflection_map_check.py` (new) tests the *ingredients* of Lemmas 7–8 directly, trajectory by trajectory: (i) the pinned model yields the cutoff Gm_i/V^2 and the recoil model $G(m_i + m_*)/V^2$ (Remark 1); (ii) the displacement $|\vec{\delta}|$ is $\mathcal{O}(b_{90})$ and r–independent; (iii) the *vector* Jacobian satisfies $\det(\partial\vec{\beta}_1^*/\partial\vec{b}) - 1 = \mathcal{O}(\eta)$, halving with η; (iv) the kick fidelity (27)–(28) holds with a small error shrinking with η, and fails if either the $(V/V_1^*)^2$ prefactor or the $b_{90,1}(V_1^*)$ shift is dropped.

8 What is proved, and what is not

Proved. (1) The exact single–body kick (8)–(9) and the self–energy identity (10): the close–encounter divergence is cut off at the *bare* $b_{90,i}$ with zero additive constant, the soft–core "$\sqrt{e}$" being cancelled by the longitudinal recoil (Lemmas 1–2); the recoil of the struck body is essential (Remark 1). (2) The longitudinal recoil never enters the anisotropy (Lemma 3). (3) The near–region reduction to the two–body problem, with the deflection map and the kick fidelity now proven (Lemmas 7–8), the core handled by measure, the near–disk contamination handled by the harmonicity cancellation (Lemma 4) and the sequential–encounter measure estimate (Lemma 5). (4) The far region gives the upper scale $r_\perp$ and the cutoff–free anisotropy 2π at straight–line accuracy, with the conditional–convergence subtlety of the self terms resolved by the circularity of the excisions (Lemma 9). (5) On assembly the intermediate scale cancels and the closed form (4)–(5), constants included, is the exact limit, including the speed average (Lemma 10); the θ– and speed–averages are exact elementary integrals.

Reduced to explicit–formula calculus. The dynamical estimates of Lemmas 7–8 rest on the quantitative regularity of the incoming leg of the Kepler flow (Lemma 11, Appendix A): boundedness, uniform on $\beta \geq \lambda b_{90}$, of the first derivatives of the flow and of the osculating–element coefficients, with constants $K(\lambda) \leq C\lambda^{-p}$. This is elementary (the flow is given by explicit conic formulas) and we indicate the computation; the constants are not optimized, and the final error rate $\mathcal{O}(\eta^c)$ inherits this non–optimality. The numerics indicate the true rate is $\mathcal{O}(\eta)$.

Established only numerically. The sharp $\mathcal{O}(\eta)$ rate of the surviving correction, its positivity in the tested configurations, and its coefficient ($\approx 1.7\,\eta$ for equal masses). None of these is needed for the theorem.

Outside scope. The frozen–binary approximation (A1); its $\mathcal{O}(\eta)$ adiabatic correction is a separate, equally vanishing effect. Relativistic and finite–size corrections (A2).

Conclusion. Within the impulse approximation, the close–encounter regularization is the literal $b_{90,i} = G(m_i + m_*)/V^2$, with no arbitrary multiplicative constant and no additive constant in the Coulomb logarithm. The companion paper's $\log \Lambda$ and the constants $-\frac{2}{3}, -\frac{8}{3}$ are the exact leading–order impulsive result; the corrections vanish with $b_{90}/r = \mathcal{O}\big((v_{\rm orb}/V)^2\big)$, the impulsive parameter itself.

A Incoming–leg regularity of the Kepler flow

Lemma 11 (Kepler incoming–leg regularity)**.** *Consider the attractive Kepler problem $\ddot{\vec{d}} = -\mu\,\vec{d}/|\vec{d}|^3$ with incoming asymptotic speed v and impact parameter $\beta \geq \lambda b$, $b \equiv \mu/v^2$, $\lambda \in (0,1]$. Measure lengths in b, times in b/v. Then there are constants $K(\lambda) \leq C\lambda^{-p}$ (some finite p, computable) such that, uniformly in $\beta/b \in [\lambda,\infty)$ and in time along the incoming leg (from the asymptote to pericenter) and the outgoing leg:*

1. *the flow map from asymptotic data $(\vec{\beta}, v\hat{u})$ to the state at any time, and the pericenter–state map and its inverse, have first derivatives bounded by $K(\lambda)$;*
2. *the transition matrices of the variational (Jacobi) equation along the orbit, from pericenter to any time, are bounded by $K(\lambda)$;*
3. *the osculating–element coefficients $\partial X/\partial\vec{u}$ for $X \in \{E, \vec{L}, \vec{A}, \vec{\beta}, \hat{u}_{\rm in}\}$, and their first derivatives with respect to the state, are bounded by $K(\lambda)$ times the appropriate dimensional scale ($|\vec{d}|/v$ for the lever–type entries).*

Proof sketch. The orbit is given explicitly by the hyperbolic–anomaly parametrization $d = a(e\cosh H - 1)$, $t\sqrt{\mu/a^3} = e\sinh H - H$, with $a = \mu/v^2 = b$ and $e = \sqrt{1+\beta^2/b^2} \geq \sqrt{1+\lambda^2}$, together with the standard in–plane position and velocity formulas; the asymptotic data enter through (e, a) and the orientation of the apse frame. Every map in the statement is a composition of these explicit algebraic/analytic functions and the inverse of the (strictly monotone) Kepler equation. Their derivatives are rational expressions in $(e, \cosh H, \sinh H)$ whose denominators are powers of $e\cosh H - 1 \geq e - 1 \geq c\lambda^2$; sup over H is finite because every expression tends, as $|H| \to \infty$, to its free–motion limit (the force decays as d^{-2}, integrably), and smooth dependence on $1/e$ up to the straight–line point $1/e = 0$ makes the family compact. Tracking the worst denominator gives $K(\lambda) \leq C\lambda^{-p}$ with p finite; we make no attempt to optimize p. For (3), the element coefficients are polynomial in $(\vec{d}, \vec{u})$ divided by powers of $V_1 = \sqrt{2E}$, $|\vec{A}| = \mu e$ and $\sqrt{e^2-1} = \beta/b$, bounded below on the incoming leg by $c\,v$, μ and λ respectively; the degenerating denominators are the powers of $e - 1 \geq c\lambda^2$ already counted above. $\square$

References

[1] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd ed. (Princeton, 2008), §3.1 (Rutherford scattering, b_{90}) and §8.2 (impulse approximation and its adiabatic correction); see also L. Spitzer, *Dynamical Evolution of Globular Clusters* (Princeton, 1987).